

Notes: Jacobson and von Schedvin (Econometrica, 2015)

Trade Credit and the Propagation of Corporate Failure: An Empirical Analysis

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1 Big picture

- **Research question.** Do trade credit links transmit corporate failures? More specifically, when a customer fails and defaults on trade credit, does the supplier face a higher probability of bankruptcy?
- **Main idea.** Supplier-customer relationships create credit chains. A bankrupt customer usually does not repay its trade creditors. Those unpaid claims can reduce supplier liquidity and net worth, while the loss of the customer also reduces future demand.
- **Empirical object.** The paper observes Swedish corporate bankruptcies, the identities of trade creditors, the identities of failed trade debtors for the baseline period, and the size of creditors' claims on failed debtors. This allows the authors to observe links in bankruptcy chains directly.
- **Main finding.** Trade debtor failures are associated with large increases in creditor failure risk. In the baseline sample, facing a trade debtor failure raises annual creditor bankruptcy risk by about 0.8 percentage points, relative to an unconditional failure rate near 1.5 percent.
- **Credit losses are economically large.** At the aggregate level, Swedish trade creditors' losses on failed customers are roughly 50 percent larger than Swedish banks' credit losses on lending to nonfinancial firms.
- **Credit channel and demand channel.** The paper argues that both mechanisms matter: suppliers are harmed by unpaid claims and by losing future sales to the failed customer.
- **Identification challenge.** The paper must separate true propagation from common shocks, endogenous matching of weak firms with weak trading partners, and reverse causality from supplier distress to customer failure.
- **Aggregate relevance.** The propagation mechanism explains a meaningful part of the aggregate bankruptcy frequency, especially during downturns. During the Swedish banking crisis, the estimated contribution reaches about 13 percent of bankruptcies.

Economic takeaway. Trade credit is not just a financing convenience. It creates a network through which firm-level failures can travel upstream from customers to suppliers and can amplify idiosyncratic shocks into aggregate bankruptcy risk.

2 Incremental contribution of the paper

- **Direct evidence on trade credit contagion.** Prior work had survey, stock-market, or indirect evidence that customer distress hurts suppliers. This paper observes actual trade credit claims in bankruptcies and directly follows failure links.
- **Creditor and debtor identities.** For 2007–2011, the data identify both sides of the trade credit link. This permits debtor fixed effects and tests comparing creditors exposed to the same failed debtor.
- **Claim size as shock severity.** The paper does not only use an indicator for debtor failure. It measures the supplier's claim on the failed debtor, scaled by supplier assets, to capture the economic size of the shock.

- **Credit losses versus demand losses.** The analysis separates two channels that are usually bundled together: the balance-sheet or liquidity effect of unpaid trade credit and the demand effect from losing a customer.
- **Careful treatment of confounds.** The paper uses combinations of time, industry, location, creditor, and debtor fixed effects; subsamples by creditor and debtor risk; relative-size tests for reverse causality; and timing robustness checks.
- **Network interpretation.** The paper distinguishes first-order and higher-order links in trade credit chains and documents stronger effects for higher-order failures, where surprises are likely to be larger.
- **Macro quantification.** Because the data cover the universe of Swedish corporate firms, the authors can ask how much of the aggregate bankruptcy frequency is attributable to the propagation mechanism.
- **Financial-stability implication.** The results imply that credit risk models ignoring supplier-customer contagion may understate both loan portfolio losses and correlation in losses across banks.

3 Data and measurement

3.1 Data sources and sample periods

The paper uses data from the Swedish credit bureau Upplysningscentralen AB. The data combine corporate bankruptcies, accounting statements, firm characteristics, credit ratings, and claims held by trade creditors on bankrupt trade debtors.

- **1992–2011 sample.** The longest sample records whether firms experience trade debtor failures. It does not always contain claim size or debtor identities.
- **1996–2011 sample.** This extended sample adds the size of claims held by trade creditors on failed trade debtors.
- **2007–2011 baseline sample.** This is the richest sample. It includes creditor identities, failed debtor identities, bankruptcy dates, and the claim sizes.
- **Firm universe.** The analysis covers the universe of Swedish corporate firms, excluding financial firms, utilities, and very small firm-years with real sales or assets below SEK 100,000.
- **Accounting data.** Annual financial statements are standardized to calendar-year timing and merged with credit ratings, industry, location, age, and balance-sheet variables.

Interpretation. The unique feature is that the paper observes actual unsecured trade credit claims in bankruptcy. This turns customer bankruptcy into an observable supplier-level shock.

3.2 Institutional setting

Swedish bankruptcy law matters for the empirical design. Once a firm is declared bankrupt, a court-appointed trustee takes control, inventories assets and liabilities, auctions assets, and pays creditors according to absolute priority.

- Trade credit is unsecured junior debt.
- Priority deviations are not allowed.
- Recovery rates for unsecured junior creditors are extremely low. Thorburn’s evidence cited in the paper puts the average recovery rate near 2 percent and the median near 0 percent.
- Because recovery is close to zero, the creditor’s claim on a bankrupt debtor is a good proxy for the creditor’s trade credit loss.

Economic implication. A customer bankruptcy is close to a permanent loss for the supplier, not just a delayed payment.

3.3 Main variables

- TCF_{it} . Indicator equal to one if firm i fails in year t .
- TDF_{it} . Indicator equal to one if firm i , as a trade creditor, experiences one or more trade debtor failures in year t .
- $Claims/Assets$. Sum of claims held by the trade creditor on failed trade debtors in year t , divided by creditor assets in $t - 1$.
- $Claims/Receivable$. Claims on failed trade debtors divided by accounts receivable.
- $12 \times Claims/Sales$. Approximate debtor-specific demand exposure. If one month of sales is lost and the debtor is a repeated customer, multiplying the claim by 12 approximates annual sales to that debtor.
- **Firm controls.** Liabilities to assets, cash and liquid assets to assets, EBIT to assets, total assets, age, credit rating, accounts receivable, tangible assets, sales/assets, and related variables.

3.4 Descriptive patterns

The descriptive statistics already suggest propagation. In the baseline period, firms with trade debtor failure exposure have higher annual bankruptcy frequency than nonexposed firms. They also issue more trade credit, hold less cash, and are larger and older on average.

The aggregate data also show that the frequency of firms experiencing debtor failures closely tracks the aggregate bankruptcy frequency. This is natural: each bankrupt debtor usually has several trade creditors, so one debtor failure can expose multiple suppliers.

3.5 Credit versus demand measures

The paper uses both credit-loss and demand-loss proxies.

- The credit-loss proxy is $Claims/Assets$: how large the unpaid claim is relative to the creditor’s balance sheet.
- The debtor-specific demand proxy is $12 \times Claims/Sales$: how large the debtor is as an implied share of the creditor’s annual sales.

- These two are imperfectly correlated. A claim can be large relative to assets but small relative to sales, or vice versa. The paper exploits this imperfect correlation to separate credit and demand channels.

4 Methodology and empirical specifications

4.1 Conceptual hypotheses

The paper organizes the empirical analysis around three hypotheses.

- **Hypothesis 1.** Trade credit issuance is positively associated with credit losses.
- **Hypothesis 2.** Trade credit losses arising in trade debtor failures impose increased trade creditor failure risk.
- **Hypothesis 3.** The trade debtor failure propagation mechanism has measurable aggregate effects.

The mechanism has two direct components. First, unpaid claims reduce creditor liquidity and net worth. Second, debtor failure removes future sales to that customer, reducing demand for the creditor's output.

4.2 Credit losses associated with trade credit issuance

To test whether trade credit issuance is associated with actual credit losses, the paper estimates Tobit models of the form:

$$\frac{Claims_{it}}{Assets_{it-1}} = \gamma_1 \frac{Receivable_{it-1}}{Assets_{it-1}} + \lambda' A_{it} + \delta' U_{it-1} + \varepsilon_{it}, \quad (1)$$

where A_{it} includes fixed effects and U_{it-1} contains firm characteristics.

- The dependent variable is nonnegative, which motivates the Tobit specification.
- The coefficient on receivables measures how trade credit issuance maps into later trade credit losses.
- The paper also estimates a logistic model in which the dependent variable is whether the firm experiences any trade debtor failure.

Interpretation. A positive coefficient means firms that issue more trade credit face larger realized trade credit losses when customers fail.

4.3 Creditor failure risk model

The core propagation regressions estimate the probability that a trade creditor fails. The paper uses a multi-period logistic model:

$$TCF_{it} \mid p_{it} \sim \text{Bernoulli}(p_{it}), \quad (2)$$

$$\log\left(\frac{p_{it}}{1-p_{it}}\right) = \beta X_{it} + \alpha_{it} + \eta' V_{it-1}, \quad (3)$$

where X_{it} is either TDF_{it} or $Claims/Assets_{it}$, α_{it} denotes fixed effects, and V_{it-1} contains creditor controls.

The paper also estimates linear probability models. The LPM is useful because the richest fixed-effect specifications would be difficult or partly unidentified in a nonlinear logit setting.

4.4 Common shocks

A customer and supplier may fail together because both are hit by the same industry-region-year shock. The paper uses two main strategies.

- **Year $\times$ industry $\times$ location fixed effects.** These compare exposed and unexposed creditors operating in the same industry, location, and year.
- **Debtor fixed effects.** These compare different creditors exposed to the same failed debtor. The identifying variation is the size of each creditor’s claim on the same debtor.

4.5 Endogenous matching

Weak customers may be matched with weak suppliers. This can create a correlation between debtor failure and creditor failure even without contagion.

- **Creditor fixed effects** compare the same creditor across years with and without debtor failure exposure.
- **Credit-rating controls** absorb time-varying observable creditor risk.
- **Low-risk subsamples** test whether the result still appears among creditors or debtors that looked relatively safe before failure.

4.6 Reverse causality

A distressed supplier might cut trade credit to a customer and cause the customer to fail. The paper argues that the data structure already reduces this concern, because a fully withdrawn claim would not appear as a creditor claim in the bankruptcy data.

The paper adds a relative-size test. If reverse causality drives the results, the effect should be strongest when creditors are much larger and therefore important credit providers to small debtors. Instead, the effect is negligible for the largest creditor-to-debtor size ratio subsample, supporting upstream contagion from debtor to creditor.

4.7 Credit-loss channel versus demand-loss channel

The paper interacts claim size with measures of financial constraints and demand dependence:

$$\theta_{it} = \beta_1 X_{it} + \beta_2 X_{it} Z_{it-1} + \beta_3 Z_{it-1} + \lambda' \alpha_{it} + \eta' V_{it-1}. \quad (4)$$

Key conditioning variables include external financing dependence, liquidity dependence, R&D intensity, leverage, cash holdings, earnings, and GDP growth.

- If the credit-loss channel matters, financially constrained and liquidity-dependent firms should be more exposed to claim losses.
- If the demand-loss channel matters, firms in industries with strong supplier-customer ties, such as R&D-intensive industries, should be more exposed.
- The paper finds evidence for both patterns.

4.8 Debtor-specific demand regressions

The paper then includes both $Claims/Assets$ and $12 \times Claims/Sales$ in the same model:

$$\theta_{it} = \beta_1 X_{it} + \beta_2 W_{it} + \lambda' \alpha_{it} + \eta' V_{it-1}, \quad (5)$$

where X_{it} contains the claim-size variables and W_{it} contains debtor-specific demand variables. Positive coefficients on both sets of variables support both the credit-loss and demand-loss channels.

4.9 Aggregate counterfactual

To evaluate aggregate relevance, the paper estimates a micro-level bankruptcy model and predicts the aggregate bankruptcy frequency twice:

- once with the trade debtor failure variable active; and
- once with the coefficient on TDF set to zero.

The difference is interpreted as the contribution of the propagation mechanism to aggregate bankruptcies in each year.

5 Identification and interpretation

5.1 What the paper can identify

The paper provides strong reduced-form evidence that trade debtor failures raise creditor bankruptcy risk. The evidence is especially persuasive because the effect survives claim-size specifications, rich fixed effects, debtor fixed effects, creditor fixed effects, timing checks, and multiple subsample tests.

5.2 Why the claim-size variable is important

An indicator for debtor failure captures the event, but not its economic severity. *Claims/Assets* is closer to the actual balance-sheet shock. The steep relationship between claim size and creditor failure risk makes the propagation interpretation more credible.

5.3 Credit channel and demand channel are both real

The paper’s conceptual contribution is to avoid treating all supplier harm as trade credit losses. The creditor loses an unpaid receivable and may also lose future customer demand. The empirical strategy uses financial constraints to identify the credit channel and supplier-customer dependence to identify the demand channel.

5.4 Common-shock concern

Industry, location, and year shocks are a serious concern because many supplier-customer pairs operate in related economic environments. The paper’s year $\times$ industry $\times$ location fixed effects and debtor fixed effects substantially reduce this concern.

5.5 Endogenous-matching concern

The authors show that the propagation pattern is not confined to obviously weak creditors or weak debtors. The result remains in low-risk subsamples and when creditor fixed effects are included.

5.6 Reverse-causality concern

Reverse causality would imply stronger effects where creditors are very large relative to debtors and therefore more able to cause debtor failure by cutting credit. The estimated effect is close to zero in this subsample, which supports the interpretation that the shock travels from debtor to creditor.

5.7 Timing interpretation

Some creditor bankruptcies occur before the debtor’s formal bankruptcy date, because distressed debtors can default on payments before legally entering bankruptcy. The paper therefore treats debtor failures close to the creditor failure date as relevant and tests alternative timing windows.

5.8 Limitations

The analysis is not a randomized experiment. Trade credit links are endogenous, claim amounts may reflect choices made before bankruptcy, and demand exposure is proxied rather than directly observed. Nevertheless, the fixed-effect structure and the battery of robustness checks make alternative explanations much harder to sustain.

6 Tables and figures

6.1 Table I and Figures 1–2: Descriptive evidence and timing

Read Table I:

- Firms with trade debtor failure exposure issue more trade credit: in the baseline period, their accounts receivable-to-assets ratio is 0.249 versus 0.153 for firms without such exposure.
- Exposed firm-years have higher bankruptcy frequency: 0.026 versus 0.014 in the baseline period.
- In the full 1992–2011 period, trade debtor failure exposure is associated with a bankruptcy frequency of 0.046 versus 0.019 for nonexposed firm-years.
- Exposed firms are larger and older on average, so the higher failure rate is not simply because they are tiny firms.

Read Figures 1 and 2:

- Figure 1 shows that aggregate bankruptcy frequency and trade debtor failure exposure move together over time, especially during the early-1990s Swedish banking crisis.
- Figure 2 shows that many creditor failures occur in the same month as the debtor failure or shortly before it, and claim sizes are especially large around the creditor failure month.

Economic message. Trade debtor failures are common, cyclical, and associated with unusually high supplier failure risk. The timing evidence is consistent with payment default preceding formal bankruptcy.

TABLE I
DESCRIPTIVE STATISTICS ON FIRM-SPECIFIC VARIABLES*

Variables	All Firms					Trade Debtor Failure Exposure		Non Trade Debtor Failure Exposure	
	Mean	Median	Std.	10th Perc.	90th Perc.	Mean	Median	Mean	Median
Panel A, 2007–2011									
<i>Accounts receivable/Assets</i>	0.158	0.092	0.182	0.000	0.432	0.249	0.223	0.153*	0.084*
<i>Accounts payable/Assets</i>	0.105	0.050	0.138	0.000	0.292	0.151	0.109	0.103*	0.047*
<i>EBIT/Assets</i>	0.067	0.064	0.238	−0.140	0.314	0.073	0.071	0.067*	0.063*
<i>Liabilities/Assets</i>	0.495	0.463	0.291	0.142	0.874	0.495	0.479	0.495	0.462*
<i>Cash holdings/Assets</i>	0.260	0.165	0.268	0.002	0.691	0.156	0.072	0.266*	0.173*
<i>Fixed assets/Assets</i>	0.315	0.193	0.315	0.003	0.843	0.280	0.174	0.318*	0.195*
<i>Assets (in SEK 1,000)</i>	33,807	2,034	994,109	362	20,889	203,020	9,410	23,750*	1,891*
<i>Sales (in SEK 1,000)</i>	27,995	2,517	575,190	372	25,563	193,056	16,846	18,185*	2,309*
<i>Sales/Assets</i>	1.941	1.548	1.736	0.186	4.036	2.275	2.051	1.922*	1.511*
<i>Number of firm employees</i>	10,522	3,000	122,832	1,000	14,000	56,995	9,000	7,760*	3,000*
<i>Firm age</i>	15.122	12.000	13.969	2.000	33.000	21.093	17.000	14.767*	12.000*
<i>Credit rating</i>	3.651	4.000	1.351	2.000	5.000	3.761	4.000	3.644*	4.000*
<i>R&D intensity</i>	0.006	0.001	0.017	0.000	0.014	0.006	0.001	0.006	0.001*
<i>External financing dependence</i>	0.097	0.135	0.346	−0.252	0.587	0.011	−0.023	0.103*	0.135*
<i>External financing dependence (W1)</i>	0.620	1.000	0.485	0.000	1.000	0.459	0.000	0.629*	1.000*
<i>Liquidity dependence</i>	0.073	0.074	0.059	0.003	0.147	0.080	0.083	0.072*	0.074*
<i>Bankruptcy (1/0)</i>	0.015	0.000	0.120	0.000	0.000	0.026	0.000	0.014*	0.000*
<i>Claims/Assets</i>	0.001	0.000	0.013	0.000	0.000	0.020	0.004		
<i>Claims/Receivable</i>	0.024	0.000	0.347	0.000	0.000	0.422	0.021		
<i>Number of firms</i>			272,003					269,721	
<i>Number of obs.</i>			1,057,935					998,587	

(Continues)

Table I, part 1: Baseline 2007–2011 firm characteristics.

TABLE I—Continued

Variables	All Firms					Trade Debtor Failure Exposure		Non Trade Debtor Failure Exposure	
	Mean	Median	Std.	10th Perc.	90th Perc.	Mean	Median	Mean	Median
Panel B, 1992–2011									
<i>Bankruptcy (1/0)</i>	0.021	0.000	0.143	0.000	0.000	0.046	0.000	0.019*	0.000*
<i>Claims/Assets[†]</i>	0.001	0.000	0.014	0.000	0.000	0.022	0.004		
<i>Claims/Receivable[‡]</i>	0.024	0.00	0.346	0.000	0.000	0.415	0.023		
<i>Number of firms</i>			445,595				113,523		438,531
<i>Number of obs.</i>			3,824,572				317,640		3,506,932

The table reports descriptive statistics for a set of firm-specific variables for the periods 2007–2011 (Panel A) and 1992–2011 (Panel B), grouped on all firm-years when the firm experienced one, or several, trade debtor failures in year t (Trade Debtor Failure Exposure), and firm-years when the firm did not experience a trade debtor failure in year t (Non Trade Debtor Failure Exposure). All firm-specific variables correspond to year $t - 1$. Variable definitions are provided in Table A-1. The star () reported next to the mean and median values for the “Non Trade Debtor Failure” group denotes that these statistics are significantly different from the “Trade Debtor failure” group’s means and medians at the 1 percent level. Differences in means are assessed using a Student’s t -test and differences in medians are assessed using the Wilcoxon–Mann–Whitney test. [†]Measured for the 1998–2011 period.

Table I, part 2: Extended 1992–2011 descriptive statistics.

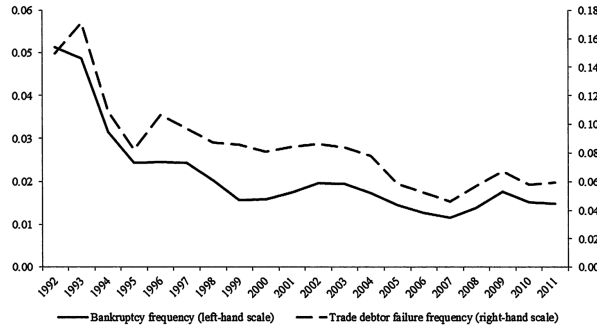


Figure 1: Bankruptcy frequency and trade debtor failure frequency.

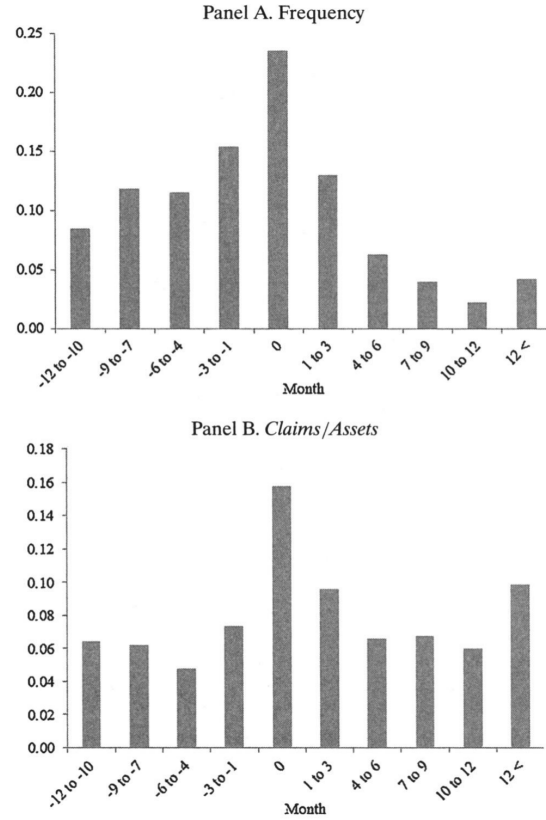


Figure 2: Timing of creditor and debtor failures, and claim sizes.

6.2 Table II: Credit losses associated with trade credit issuance

Read:

- Table II estimates Tobit and logit models linking issued trade credit to later trade credit losses.
- The coefficient on receivables-to-assets is positive and highly significant across specifications.
- In the extended 1996–2011 period, a one-unit increase in receivables-to-assets is associated with a 0.077 increase in claims-to-assets.
- Interpreted in dollar terms, a \$100 increase in outstanding trade credit is associated with about \$7–\$8 more in annual credit losses.
- A logit model also shows that more trade credit issuance increases the probability of experiencing a debtor failure.

Economic message. Trade credit issuance is risky. Suppliers that extend more credit to customers experience meaningfully larger losses when customers fail.

6.3 Tables III–IV and Figure 3: Creditor failure risk

Read Table III:

- The dependent variable is creditor bankruptcy, TCF .
- The TDF indicator is positive and significant in the baseline and extended samples.
- The estimated marginal effect is about 0.008 in the baseline period and 0.012 in the extended period.
- The result remains after adding year, industry, location, creditor, and interacted fixed effects.
- The effect is close to zero for cases where creditors are very large relative to debtors, which weakens the reverse-causality explanation.
- Higher-order debtor failures have stronger effects than first-order failures, suggesting that surprise and network position matter.

Read Figure 3 and Table IV:

- Figure 3 shows a steep positive relation between $Claims/Assets$ and realized creditor bankruptcy rates.
- Table IV shows that claim size predicts creditor failure in logit, quadratic logit, LPM, creditor fixed-effect, and debtor fixed-effect specifications.
- The benchmark model implies that larger losses strongly increase bankruptcy risk.
- The debtor fixed-effect result compares creditors of the same failed debtor, making the claim-size result difficult to explain by debtor-level shocks alone.

Economic message. It is not merely being linked to a failed customer that matters. The size of the unpaid claim is central. Larger losses impose larger bankruptcy risk on suppliers.

TABLE II
CREDIT LOSSES ASSOCIATED WITH TRADE CREDIT ISSUANCE^a

Variables	Dependent Variables				
	Claims/Assets		Claims/Sales	TDF (0/1)	
	(I) Coef.	(II) Coef.	(IV) Coef.	(V) $d\theta/dx$	(VI) $d\theta/dx$
Receivable/Assets	0.071*** (57.1)	0.077*** (113.4)	0.060*** (102.8)	0.142*** (117.8)	
Receivable/Assets; 5–10 p.					
Receivable/Assets; 10–15 p.					
Receivable/Assets; 15–25 p.					
Receivable/Assets; 25–35 p.					
Receivable/Assets; 35 p.					
Firm controls	Yes	Yes	Yes	Yes	Yes
Age FE	Yes	Yes	Yes	Yes	Yes
Credit rating FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes	Yes
Location FE	Yes	Yes	Yes	Yes	Yes
Model	Tobit	Tobit	Tobit	Tobit	Logistic
Sample period	2007–2011	1996–2011	1996–2011	1996–2011	1992–2011
Pseudo- R^2	0.367	0.401	0.418	0.448	0.188
F/χ^2	59***	221***	218***	169***	121,981***
Area under ROC curve	—	—	—	—	0.806
Number of obs.	1,057,935	3,141,711	3,141,711	3,141,711	3,824,572

^aThe table reports coefficients from Tobit regressions (Equation (1)) estimating the credit losses associated with trade credit issuance and average marginal effects from a logistic regression estimating the likelihood that a firm experiences one, or more, trade debtor bankruptcies in year t , for the periods 2007–2011, 1996–2011, and 1992–2011. The dependent variable in the Tobit regressions is the size of the claim that the trade creditor has on a bankrupt trade debtor at time t , scaled by total (creditor) assets at time $t-1$, or by total (creditor) sales at time $t-1$; and the dependent variable in the logistic regression, TDF , indicates whether or not a firm experienced one, or more, trade debtor bankruptcies in year t . For the Tobit estimation, we apply a lower truncation limit equal to zero. If a trade creditor experiences multiple debtor failures in a year, then we enter the sum of the claims into the dependent variable. Firm controls include the natural logarithm of total assets, tangible assets over assets, external financing dependence, and $EBIT$ over assets. Quadratic and cubic terms of the firm controls are included. The firm-specific variables are described in Table A-I. Industry-fixed effects correspond to one-digit SNI codes, location is determined at the county level (Swedish län, 21 regions), and credit rating is the rating assigned by the credit bureau UC. The pseudo- R^2 values are calculated according to McFadden (1973). The F -test evaluates whether the parameters associated with the trade debtor failure and the firm-specific controls are jointly equal to zero. The χ^2 -statistic refers to a Wald test of the null that all parameters, except the intercept, are jointly equal to zero. The ROC-measure refers to the receiving operating characteristic; gauging in-sample, overall predictive ability. t -values, calculated using robust standard errors, clustered at the firm-level, are reported within parentheses. ***, **, * denote statistically distinct from 0 at the 1, 5, and 10 percent level, respectively.

Table II: Credit losses associated with trade credit issuance.

TABLE III—Continued

Variables	Dependent Variable: TCF (0/1)						
	(I) $d\theta/dx$	(II) Coef.	(III) Coef.	(IV) Coef.	(V) $d\theta/dx$	(VI) $d\theta/dx$	(VII) $d\theta/dx$
Panel B. Extended period							
TDF (0/1)	0.012*** (55.1)	0.018*** (47.4)	0.018*** (47.7)	0.013*** (36.6)			
Pseudo- R^2 / R^2	0.177	0.045	0.081	0.467			
F/χ^2	104,047***	4,093***	3,830**	3,615***			
Area under ROC curve	0.844	—	—	—			
Number of obs.	3,824,572	3,824,572	3,824,572	3,824,572			
Firm controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Credit rating FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Location FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year $\times$ Industry $\times$ Location FE			Yes				
Credit rating FE				Yes			
Model	Logistic	LPM	LPM	LPM	Logistic	Logistic	Logis

^aThe table reports results from regressions estimating the likelihood that a firm fails as an outcome of facing a trade debtor bankruptcy (Equation (2)) where the TDF variable is included to measure the exposure of trade debtor failures). Average marginal effects are reported for the logistic models. Panels A and B report results obtained for the base period 2007–2011, and the extended period 1992–2011. The dependent variable, TCF , indicates whether a firm is bankrupt or not in year t . TDF is an indicator variable taking the value 1 if a firm experienced a trade debtor bankruptcy, and zero otherwise, in year t . $TDF^{relative\ large\ creditor}$ corresponds to debtor failures for which the creditor belongs to the top decile in the relative creditor-debtor size distribution, measured by creditor assets over debtor assets. $TDF^{relative\ small\ creditor}$ relates to trade debtor failures where the debtor did not experience a debtor failure prior to his failure. $TDF^{prior\ creditor}$ corresponds to debtor failures where the debtor in turn also experienced a debtor failure prior to fail. $TDF^{prior\ debtor}$ corresponds to the fractions of firms in the creditor's debtor industries that experienced a trade debtor failure. All firm-specific variables correspond to year $t-1$. The firm-specific variables are described in Table A-I. Credit rating fixed effects are estimated using ratings assigned by the credit bureau UC. Industry-fixed effects are based on one-digit SNI codes, and location is determined at the county level (Swedish län, 21 regions). Industry-fixed effects in Column (II) are based on two-digit SNI codes. $\bar{P}$ pseudo- R^2 values are calculated according to McFadden (1973). The F -test evaluates whether the parameters associated with the TDF variables and the firm-specific controls are jointly equal to zero. The χ^2 -statistic refers to a Wald test of the null that all parameters, except the intercept, are jointly equal to zero. The ROC-measure refers to the receiving operating characteristic; gauging in-sample, overall predictive ability. t -values are calculated using robust standard errors, clustered at the creditor-level, and are reported with parentheses. ***, **, * denote statistically distinct from 0 at the 1, 5, and 10 percent level, respectively.

Table III, part 2: Extended-period estimates and fixed-effect controls.

TABLE III
CREDITOR FAILURE RISK ASSOCIATED WITH DEBTOR FAILURES^a

Variables	Dependent Variable: TCF (0/1)						
	(I) $d\theta/dx$	(II) Coef.	(III) Coef.	(IV) Coef.	(V) $d\theta/dx$	(VI) $d\theta/dx$	(VII) $d\theta/dx$
Panel A. Baseline period							
TDF (0/1)	0.008*** (18.9)	0.011*** (16.5)	0.010*** (16.0)	0.010*** (12.4)	0.008*** (19.1)		0.009** (18.1)
$TDF^{relative\ large\ creditor}$ (0/1)					-0.001 (-0.2)		
$TDF^{prior\ debtor}$ (0/1)						0.007** (14.3)	
$TDF^{prior\ creditor}$ (0/1)						0.015*** (16.5)	
$TDF^{prior\ debtor}$ (0/1)							0.057** (4.5)
Pseudo- R^2 / R^2	0.227	0.061	0.104	0.591	0.227	0.227	0.235
F/χ^2	28,205***	774***	751***	318***	28,224***	28,221***	23,777*
Area under ROC curve	0.881	—	—	—	0.881	0.881	0.885
Number of obs.	1,057,935	1,057,935	1,057,935	1,057,935	1,057,935	1,057,935	825,46

(Continue)

Table III, part 1: Debtor failure exposure and creditor failure risk.

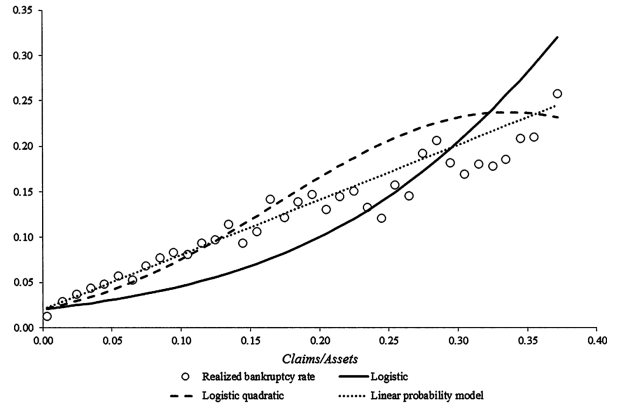


Figure 3: Trade credit losses and creditor bankruptcy risk.

TABLE IV
CREDITOR FAILURE RISK ASSOCIATED WITH TRADE CREDIT LOSSES^a

Variables	Dependent Variable: <i>TCF</i> (0/1)				
	(I)	(II)	(III)	(IV)	(V)
	$d\theta/dx$	Benchmark $d\theta/dx$	Coef.	Coef.	Coef.
<i>Panel A. Baseline period</i>					
<i>Claims/Assets</i>	0.100*** (28.6)	0.200*** (17.5)	0.483*** (18.3)	0.367*** (12.7)	0.242*** (7.0)
Pseudo- R^2/R^2	0.230	0.231	0.106	0.592	0.595
F/χ^2	31,608***	31,878***	766***	320***	16***
Area under <i>ROC</i> curve	0.882	0.883	—	—	—
Number of obs.	1,057,935	1,057,935	1,057,935	1,057,935	40,269
<i>Panel B. Extended period</i>					
<i>Claims/Assets</i>	0.120*** (59.3)	0.229*** (35.9)	0.547*** (39.0)	0.430*** (30.3)	
Pseudo- R^2/R^2	0.204	0.204	0.080	0.433	
F/χ^2	91,859***	92,966***	2,534***	2,599***	
Area under <i>ROC</i> curve	0.863	0.864	—	—	
Number of obs.	3,141,711	3,141,711	3,141,711	3,141,711	
Firm controls	Yes	Yes	Yes	Yes	Yes
Credit rating FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes		Yes	Yes
Industry FE	Yes	Yes		Yes	Yes
Location FE	Yes	Yes		Yes	Yes
Year $\times$ Industry $\times$ Location FE			Yes		
Creditor FE				Yes	
Debtor FE					Yes
<i>Claims/Assets</i> ² Included		Yes			
Model	Logistic	Logistic	LPM	LPM	LPM

^aThe table reports results from regressions estimating the creditor bankruptcy risk associated with trade credit losses (Equation (2)) where the *Claims/Assets* variable is included to measure the exposure of trade debtor failures). Average marginal effects are reported for the logistic models. Panel A reports results obtained for the baseline period 2007–2011, and Panel B reports results obtained from the extended period 1996–2011. The dependent variable, *TCF*, indicates whether a firm is bankrupt or not in year t . *Claims/Assets* is the sum of the claims that the trade creditor has on bankrupt trade debtors at time t to total (creditor) assets at time $t - 1$. Firm controls are: liabilities to assets, cash and liquid assets to assets, *EBIT* to assets, logarithm of total assets, and logarithm of one plus the firm age (both linearly and squared). All firm-specific variables correspond to year $t - 1$. The firm-specific variables are described in Table A-I. Credit rating-fixed effects are estimated using ratings assigned by the Swedish credit bureau UC. Industry-fixed effects are based on one-digit SNI codes, and location is determined at the county level (Swedish län, 21 regions). Industry-fixed effects in Column (III) are based on two-digit SNI codes. The pseudo- R^2 values are calculated according to McFadden (1973). The F -test evaluates whether the parameters associated with the trade debtor failure and the firm-specific controls are jointly equal to zero. The χ^2 -statistic refers to a Wald test of the null that all parameters, except the intercept, are jointly equal to zero. The *ROC*-measure refers to the receiving operating characteristic; gauging in-sample, overall predictive ability. t -values are calculated using robust standard errors, clustered at the creditor-level (except for the model with debtor-fixed effects where clustering is done at the debtor-level), and are reported within parentheses. ***, **, * denote statistically distinct from 0 at the 1, 5, and 10 percent level, respectively.

Table IV: Creditor failure risk associated with trade credit losses.

6.4 Tables V–VI: Credit losses, demand losses, and heterogeneity

Read Table V:

- The claim-size effect is stronger for creditors in industries dependent on external finance.
- The effect is stronger for liquidity-dependent creditors.
- The effect is stronger for highly leveraged creditors and weaker for firms with high cash holdings or high earnings.
- The effect is also stronger in R&D-intensive industries, consistent with the demand channel: specialized supplier-customer relationships make lost demand more damaging.
- The effect is larger in downturns, consistent with tighter external finance and weaker aggregate demand.

Read Table VI:

- Panel A focuses on cases where debtor-specific demand is small. The claim-size effect remains strong even when the implied customer share of creditor sales is low.
- Panel B includes both *Claims/Assets* and $12 \times \text{Claims/Sales}$.
- Both variables are positive and significant in many specifications.
- Financial-dependence interactions mainly strengthen the credit-loss effect, while R&D interactions strengthen the demand-loss effect.

Economic message. The propagation mechanism is not one-dimensional. Credit losses matter because they reduce liquidity and net worth; demand losses matter because failed customers stop buying from suppliers.

TABLE V
CROSS-SECTIONAL DETERMINANTS OF CREDITOR FAILURE RISK ASSOCIATED WITH TRADE CREDIT LOSSES*

	Conditional Marginal Effects of Claims/Assets						
	(I) External Financing Dependence	(II) External Financing Dependence (0/1)	(III) Liquidity Dependence	(IV) R&D Intensity	(V) Liabilities/ Assets	(VI) Cash Holdings/ Assets	(VII) Earnings/ Assets
Panel A. Baseline period							
10th perc. (dummy = 0)	0.180*** (9.5)	0.160*** (10.9)	0.130*** (6.6)	0.192*** (15.7)	0.161*** (13.2)	0.300*** (16.1)	0.217*** (16.9)
90th perc. (dummy = 1)	0.221*** (11.6)	0.251*** (14.0)	0.295*** (12.4)	0.270*** (7.6)	0.292*** (17.9)	-0.059 (-1.6)	0.159*** (14.5)
p-value	0.177	0.000	0.000	0.037	0.000	0.000	0.000
Panel B. Extended period							
10th perc. (dummy = 0)	0.216*** (22.1)	0.189*** (24.1)	0.179*** (17.3)	0.218*** (32.7)	0.169*** (25.5)	0.526*** (31.9)	0.250*** (34.4)
90th perc. (dummy = 1)	0.241*** (23.3)	0.264*** (27.3)	0.300*** (21.9)	0.316*** (16.7)	0.331*** (35.6)	-0.020 (-1.1)	0.164*** (25.5)
p-value	0.096	0.000	0.000	0.000	0.000	0.000	0.000

*The table reports conditional average marginal effects. Panel A reports results obtained for the baseline period 2007–2011, and Panel B reports results obtained from the extended period 1996–2011. The conditional average marginal effects are obtained from Equation (3) where each factor, one at a time, is included both linearly and through interactions with *Claims/Assets* and *Claims/Assets*². The p-value refers to a test of the null hypothesis that the marginal effect at the 10th percentile is equal to that at the 90th percentile. ***, **, * denote statistically distinct from 0 at the 1, 5, and 10 percent level, respectively.

Table V: Cross-sectional determinants of the claim-size effect.

TABLE VI
CONTROLLING FOR DEBTOR-SPECIFIC DEMAND*

Panel A	Baseline Period		Extended Period	
	(I) Mean of $12 \times \text{Claims/Sales}$	(II) Conditional Marginal Effects of Claims/Assets	(III) Mean of $12 \times \text{Claims/Sales}$	(IV) Conditional Margi Effects of Claims/A
Row	Threshold (\$)			
(1)	≤ 100 percent	0.183	0.200*** (17.5)	0.229*** (35.9)
(2)	< 25 percent	0.040	0.288*** (6.1)	0.228*** (11.5)
(3)	< 10 percent	0.023	0.295*** (6.0)	0.266*** (6.8)
(4)	< 5 percent	0.015	0.361*** (3.6)	0.208*** (2.8)

(Continued)

Table VI, part 1: Debtor-specific demand thresholds.

TABLE VI—Continued

Panel B		Baseline Period				Extended Period			
Row	Variables	$d\theta/dx$	10th Percentile (Dummy = 0)	90th Percentile (Dummy = 1)	p-Value	$d\theta/dx$	10th Percentile (Dummy = 0)	90th Percentile (Dummy = 1)	p-Value
A. Overall effects									
(1)	<i>Claims/Assets</i>	0.129*** (7.9)				0.155*** (15.9)			
(2)	$12 \times \text{Claims/Sales}$	0.011*** (5.9)				0.011*** (9.8)			
B. External financing dependence (0/1)									
(3)	<i>Claims/Assets</i>	0.089*** (4.2)	0.179*** (6.7)	0.007		0.130*** (10.5)	0.185*** (12.0)	0.005	
(4)	$12 \times \text{Claims/Sales}$	0.013*** (4.7)	0.009*** (3.3)	0.238		0.011*** (7.2)	0.011*** (6.9)	0.758	
C. Liquidity dependence									
(5)	<i>Claims/Assets</i>	0.084*** (3.0)	0.192*** (6.1)	0.029		0.133*** (8.5)	0.187*** (9.5)	0.066	
(6)	$12 \times \text{Claims/Sales}$	0.009*** (2.9)	0.013*** (4.1)	0.463		0.008*** (4.3)	0.015*** (7.3)	0.033	
D. R&D Intensity									
(7)	<i>Claims/Assets</i>	0.128*** (7.3)	0.133*** (2.8)	0.918		0.149*** (14.6)	0.195*** (6.5)	0.139	
(8)	$12 \times \text{Claims/Sales}$	0.010*** (5.0)	0.018*** (3.6)	0.136		0.010*** (8.8)	0.018*** (5.1)	0.045	
E. R&D Intensity (0/1)									
(9)	<i>Claims/Assets</i>	0.129*** (7.3)	0.123*** (2.5)	0.903		0.157*** (15.1)	0.117*** (4.1)	0.187	
(10)	$12 \times \text{Claims/Sales}$	0.009*** (4.6)	0.022*** (4.3)	0.022		0.010*** (8.0)	0.021*** (6.4)	0.001	

*Panel A: Columns (I) and (III) report the means of debtor-specific demand, $12 \times \text{Claims/Sales}$, which is less than or equal to 100, and less than 25, 10, and 5 percent of creditor's total sales, Rows (1)–(4), respectively. Columns (II) and (IV) report conditional marginal effects obtained from Equation (4) for which a variable $D(12 \times \text{Claims/Sales} < 5\%)$ enters linearly and through an interaction term with *Claims/Assets* and *Claims/Assets*². $D(12 \times \text{Claims/Sales} < 5\%)$ is a dummy variable taking the value 1 if $12 \times \text{Claims/Sales}$ is below the cut-off of 5 percent of creditor's total sales, and zero otherwise. The marginal effects correspond to claims for which $D(12 \times \text{Claims/Sales} < 5\%) = 1$. Panel B: Rows (1) and (2) report average marginal effects obtained from estimations of Equation (5). Rows (3) to (10) report results from augmented versions of the model reported in Rows (1) and (2), where the variables corresponding to *Claims/Assets* and $12 \times \text{Claims/Sales}$ are, in turn, interacted with our measures of financing dependence, liquidity dependence, and R&D intensity. A linear term of the propagation factors is included as well. Variable definitions are provided in Table A-I. The p-value refers to a test of the null hypothesis that the marginal effect at the 10th percentile is equal to that at the 90th percentile. ***, **, * denote statistically distinct from 0 at the 1, 5, and 10 percent level, respectively.

Table VI, part 2: Credit-loss and demand-loss channels jointly estimated.

6.5 Figure 4: Aggregate relevance

Read:

- Figure 4 uses the extended-period bankruptcy model to estimate the share of aggregate bankruptcies attributable to trade debtor failure exposure.
- The contribution is highest during the Swedish banking crisis of the early 1990s.
- In 1993, the propagation mechanism explains around 13 percent of the aggregate bankruptcy frequency.

- In more normal years, the contribution is closer to about 5 percent.

Economic message. Trade credit chains do not only matter in micro regressions. They explain a measurable fraction of aggregate bankruptcy risk, especially in bad times.

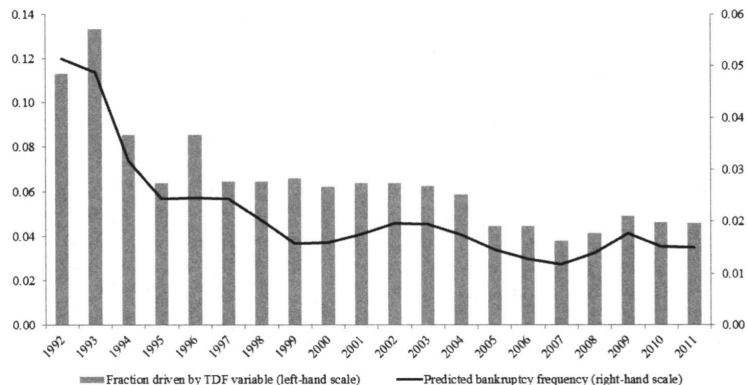


Figure 4: Contribution of trade debtor failure exposure to aggregate bankruptcy frequency.

6.6 Appendix tables: Variable definitions and robustness

Read Table A-I:

- The appendix defines the key failure, exposure, claim, balance-sheet, financial-dependence, liquidity-dependence, and R&D variables.
- It clarifies the construction of *TDF*, first-order and higher-order failures, indirect exposure, and the claim-size variables.

Read Tables A-II and A-III:

- Table A-II reports robustness checks including splines, Altman-style z-score variables, alternative event windows, more detailed industry fixed effects, low-risk creditor subsamples, higher-order debtor failures, low-risk debtor subsamples, and debtor-specific demand controls.
- The claim-size coefficient remains positive and significant across these robustness tests.
- Table A-III estimates quarterly forecasting models. Trade credit losses predict creditor failure up to several quarters ahead.

Economic message. The appendix strengthens the case that the baseline effect is not an artifact of one timing rule, one set of controls, one industry classification, or one frequency of observation.

APPENDIX

TABLE A-I
VARIABLE NAMES AND DEFINITIONS^a

Variable Names	Definitions
<i>TCF</i> (0/1)	Takes the value 1 if a trade creditor fails in year t , and zero otherwise
<i>TDF</i> (0/1)	Takes the value 1 if a trade creditor experiences one or more trade debtor failures in year t , and zero otherwise. If $TCF = 1$, then TDF takes the value 1 if the creditor experiences a debtor failure in the same year or in the twelve months prior to the creditor failure
<i>TDF</i> ^{relative large creditor} (0/1)	Corresponds to debtor failures for which the creditor belongs to the top decile of the relative creditor-debtor size distribution, measured by total assets
<i>TDF</i> ^{first-order} (0/1)	Relates to trade debtor failures where the debtor did not experience a debtor failure prior to his failure
<i>TDF</i> ^{higher-order} (0/1)	Corresponds to debtors failures where the debtor itself experienced a debtor failure prior to its failure
<i>TDF</i> ^{indirect} (0/1)	Corresponds to the fraction of firms in the creditor's debtor industries that experience a trade debtor failure (see definition in Section 4.2)

(Continues)

Table A-I, part 1: Variable definitions.

Variable Names	Definitions
<i>Claims</i>	The sum of the claims held on failed trade debtors in year t . The sum of the claims are calculated for all events underlying the <i>TDF</i> variable
<i>Accounts receivable</i>	The amount of issued trade credit
<i>Accounts payable</i>	The amount of received trade credit
<i>EBIT</i>	Earnings before interest and taxes
<i>Total liabilities</i>	Total liabilities excluding accounts payable
<i>Cash holdings</i>	Cash and liquid assets
<i>Tangible assets</i>	Property, plants, and equipment
<i>Assets</i>	Total assets
<i>Sales</i>	Total sales
<i>Sales/Assets</i>	Total sales over total assets; asset turnover ratio
<i>Number of firm employees</i>	Owner manager plus the number of employees
<i>Firm age</i>	Number of years since first registered as a corporation
<i>Credit rating</i>	Spans between 1 (high risk) and 5 (low risk) and is the rating assigned by the Swedish credit bureau, UC
<i>External financing dependence</i>	Is calculated according to Rajan and Zingales (1998) for U.S. public firms and assigned to the firms in the sample
<i>External financing dependence</i> (0/1)	Takes the value 1 for firms belonging to an industry characterized by a positive external financing dependence, and zero otherwise
<i>Liquidity dependence</i>	Is calculated according to Raddatz (2006) for U.S. public firms and assigned to the firms in the sample
<i>R&D Intensity</i>	The total amount of <i>R&D</i> expenditures scaled by the total amount of net sales for firms with more than 10 employees, calculated on a two-digit industry level
<i>R&D Intensity</i> (0/1)	Takes the value 1 for firms belonging to the top 25 percent of industries with the highest <i>R&D</i> intensity, and zero otherwise

^aThis table provides definitions for the variables used in the analysis.

Table A-I, part 2: Continued variable definitions.

TABLE A-II ROBUSTNESS ^a									
Variables	Dependent Variable: <i>TCF</i> (0/1)								
	(I) Coef.	(II) Coef.	(III) Coef.	(IV) Coef.	(V) Coef.	(VI) Coef.	(VII) Coef.	(VIII) Coef.	(IX) Coef.
<i>Claims/Assets</i>	0.259*** (7.0)	0.238*** (6.9)	0.231*** (7.0)	0.313*** (7.9)	0.402*** (16.1)	0.299*** (8.8)	0.175*** (5.0)	0.309*** (3.1)	0.314*** (4.9)
$12 \times \text{Claims/Sales}$									
R^2	0.597	0.596	0.631	0.471	0.264	0.322	0.604	0.506	0.652
F -test	4.9***	19.4***	13.6***	23.8***	34.8***	26.8***	9.5***	6.1***	5.6***
Number of obs.	40,269	40,269	40,120	40,385	1,057,935	32,743	34,497	5,397	10,540

(Cont.)

Table A-II, part 1: Robustness checks.

Variables	Dependent Variable: <i>TCF</i> (0/1)								
	(I) Coef.	(II) Coef.	(III) Coef.	(IV) Coef.	(V) Coef.	(VI) Coef.	(VII) Coef.	(VIII) Coef.	(IX) Coef.
Firm controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
<i>Z</i> -score variables	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Location FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year $\times$ Ind. FE					Yes				
Year $\times$ Loc. FE					Yes				
Year $\times$ Ind. $\times$ Loc. FE					Yes				
Debtor FE	Yes	Yes	Yes	Yes	5-digit	2-digit	Yes	Yes	Yes
SNI Level. ind. classification	1-digit	1-digit	1-digit	1-digit	Annual	Annual	1-digit	1-digit	1-digit
Mechanism window	Annual	Annual	Monthly	Monthly	Annual	Annual	Annual	Annual	Annual
Sample	All	All	All	All	All	All	Low-risk	Low-risk	Low-risk
Model	Spline	LPM	LPM	LPM	LPM	LPM	LPM	LPM	LPM

^aThe table reports results from regressions estimating the creditor bankruptcy risk associated with trade credit losses (Equation (2) where the *Claims/Assets* variable is used to measure the exposure of trade debtor failures), for a data set on a quarterly frequency covering the period 2007–2011. Column (I) reports results from a spline version of the debtor-firm effects model reported in (V). Table V. In the additive spline model, we include knots at the 20th, 40th, 60th, and 80th percentiles of each control variable (the model corresponds to a LPM $\times$ Equation (5) in Giannini et al. (2014)). Column (II) reports results for a debtor-firm effects model including additional control variables related to the asset turnover ratio working capital. Column (III) reports results from a model in which debtor failures that occur after the associated creditor failure are omitted. Column (IV) reports a model in which debtor failures that are observed at any point in time after the associated creditor failure are assigned to the creditor failure year. Column (V) reports from a version of the model in Column (III), Table V, where the industry is classified at the five-digit level. Column (VI) reports results from a model that includes the firm for: year $\times$ creditor industry, year $\times$ location, and year $\times$ debtor industry, where industry is classified at the two-digit level. Column (VII) reports results from a debtor-firm model estimated on a subsample of creditors with a credit rating of 3, or better (corresponding to a yearly default risk of 3.04 percent and below). Column (VIII) reports from a debtor-firm effect model estimated on a subsample of higher-order debtor failures (debtors that experienced a debtor failure prior to their own failure). Column (IX) reports results from a debtor-firm effect model estimated on the subsample of debtors with a credit rating of 3, or better, just prior to failure. Column (X) reports results from a debtor-firm effect model including a variable related to the debtor-specific demand, $12 \times \text{Claims/Sales}$. All firm-specific variables correspond to year $t - 1$. The firm variables are described in Table A-I. The F -test evaluates whether the parameters associated with the trade debtor failure and the firm-specific controls are jointly equal to zero. t -values calculated on robust standard errors, clustered at the debtor-level (except for the models in Columns (V) and (VI) where clustering is done at the creditor-firm level reported within parentheses). ***, **, * denote statistically distinct from 0 at the 1, 5, and 10 percent level, respectively.

Table A-II, part 2: Continued robustness checks.

TABLE A-III
FORECASTING MODELS ESTIMATED ON A QUARTERLY FREQUENCY^a

Variable	$p(TCF_{t+j} = 1 TCF_{t+j-1} = 0)$						
	(I) $j = 0$ $d\theta/dx$	(II) $j = 1$ $d\theta/dx$	(III) $j = 2$ $d\theta/dx$	(IV) $j = 3$ $d\theta/dx$	(V) $j = 4$ $d\theta/dx$	(VI) $j = 6$ $d\theta/dx$	(VII) $j = 8$ $d\theta/dx$
<i>Claims/Assets</i>	0.030*** (8.7)	0.035*** (8.6)	0.032*** (6.0)	0.023*** (3.4)	0.024*** (3.0)	0.032*** (3.8)	0.006 (0.6)
Pseudo- R^2	0.159	0.158	0.151	0.146	0.137	0.128	0.115
χ^2	14,703***	19,193***	21,458***	20,512***	18,718***	16,777***	13,323
Area under ROC curve	0.884	0.877	0.866	0.857	0.847	0.836	0.822
Number of obs.	3,954,646	3,832,910	3,676,472	3,462,862	3,249,373	2,811,993	2,381,0
Period	2007q1–2011q4	2007q1–2011q3	2007q1–2011q2	2007q1–2011q1	2007q1–2010q4	2007q1–2010q2	2007q1–2010q1
Firm controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Credit rating FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year-quarter FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Location FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
<i>Claims/Assets</i> ² included	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Model	Logistic	Logistic	Logistic	Logistic	Logistic	Logistic	Logist

^aThe table reports average marginal effects from regressions estimating the creditor bankruptcy risk associated with trade credit losses (Equation (2) where *Claims* and *Claims/Assets*² are included to measure the exposure of trade debtor failures), for a data set on a quarterly frequency covering the period 2007–2011. The models are the firm-failure probability in quarter $t + j$ conditional on survival in $t + j - 1$. All firm-specific variables correspond to quarter $t - 4$. The firm-specific variables are described in Table A-I. The pseudo- R^2 values are calculated according to McFadden (1973). The χ^2 -statistic refers to a Wald test of the null that all parameters, except the intercept jointly equal to zero. t -values calculated on robust standard errors, clustered at the firm-level, are reported within parentheses. The ROC-measure refers to the receiving operating characteristic; gauging in-sample, overall predictive ability. ***, **, * denote statistically distinct from 0 at the 1, 5, and 10 percent level, respectively.

Table A-III: Quarterly forecasting models.

7 Short summary

- The paper studies whether trade credit links propagate corporate failure from customers to suppliers.
- It uses Swedish universe-level corporate data and bankruptcy records.
- The key data advantage is observing creditor claims on failed trade debtors.
- Trade credit issuance is associated with economically important credit losses.
- Firms exposed to a trade debtor failure have much higher subsequent bankruptcy risk.
- The effect is stronger when the creditor’s claim on the failed debtor is larger relative to creditor assets.
- The result survives controls for common shocks, endogenous matching, creditor fixed effects, debtor fixed effects, and timing choices.
- The propagation mechanism is not driven only by credit losses. Demand losses from losing a customer also matter.
- Financially constrained, liquidity-dependent, highly levered, cash-poor, and low-earnings creditors are more exposed to debtor failure shocks.
- R&D-intensive industries show stronger effects, consistent with important supplier-customer relationships and demand losses.
- Higher-order failures in credit chains have larger effects than first-order failures.
- At the aggregate level, the propagation mechanism explains a meaningful share of Swedish bankruptcies, especially during downturns.

8 Conclusion

8.1 Core conclusion

Jacobson and von Schedvin show that trade credit chains propagate corporate failure. When a customer goes bankrupt, the supplier faces a higher probability of bankruptcy, and the effect increases with the size of the unpaid trade credit claim.

8.2 Economic intuition

The mechanism is simple but powerful. Suppliers finance customers through accounts receivable. When customers fail, suppliers lose cash inflows and suffer a balance-sheet hit. At the same time, the supplier may lose future demand from that customer. These shocks are especially damaging when suppliers are financially constrained or when customer relationships are hard to replace.

8.3 Incremental contribution revisited

The paper moves the literature beyond indirect evidence by observing actual trade credit claims in bankruptcy. It directly measures the creditor-debtor link, the size of the loss, and the subsequent creditor failure outcome. It also shows that the micro-level propagation mechanism has aggregate relevance.

8.4 Implications

The findings matter for corporate finance, macro-finance, and financial stability. Firms are linked not only through bank loans or ownership networks, but also through routine commercial credit. A bank lending to many firms may face correlated losses if one customer failure triggers supplier failures throughout the trade credit network.

8.5 Limitations

The setting is Sweden, and the empirical design is observational. The demand channel is proxied rather than directly observed, and trade credit relationships are endogenous. Still, the richness of the data and the breadth of fixed-effect and robustness tests make the failure propagation interpretation highly credible.

8.6 Takeaway

The enduring takeaway is that trade credit is a bankruptcy propagation channel. Customer failures impose credit and demand losses on suppliers, raise supplier bankruptcy risk, and help explain why firm failures cluster over time.